Exercise Sheet

Lecture 14

Operators on Hilbert Space

Throughout, $\mathcal{H}$ is complex, $\mathcal{B}(\mathcal{H})$ denotes bounded operators, and an unbounded operator includes its specified dense domain. $\mathrm{spec}(T)$ is the spectrum; $T^\dagger$ the adjoint; inner products follow the physics convention. Quantum questions distinguish the finite ideal-projective model from the general density-operator/POVM-instrument/channel framework. A $\star$ marks a harder problem.

Core tutorial route

Work **14.1, 14.7, 14.9, 14.10, 14.17, 14.19, 14.26, and 14.27** in that order (about 90–105 minutes before discussion). This route covers norm and adjoint; point/continuous/residual comparison; normalized sequences proving both position and momentum unbounded; common-domain CCR plus the finite trace obstruction; boundary conditions and symmetric versus self-adjoint reasoning; the qualified compact theorem and expansion; and domain-aware PVM, Stone, and quantum-model interpretation.

Additional practice: 14.2–14.6 and 14.8. **Bridge:** 14.20–14.25 (finite ideal measurement and dynamics). **Extension:** 14.11–14.16 and 14.18 (polynomial commutators and ladder algebra). Bridge and extension tasks do not count toward core progress.

Bounded operators, adjoints, spectrum

Exercise 14.1 (computational) Find the operator norm and the adjoint of multiplication by $m(x) = e^{ix}$ on $L^2[0, 2\pi]$.

Answer. $\|M\| = 1$; adjoint is multiplication by e^{-ix}; M is unitary.

Exercise 14.2 (computational) The diagonal operator $De_n = \frac{n}{n+1}e_n$ on ℓ^2. Find its point spectrum and full spectrum.

Answer. Point spectrum $\{\frac{n}{n+1} : n \geq 1\}$; $\mathrm{spec}(D) = \{\frac{n}{n+1}\} \cup \{1\}$.

Exercise 14.3 (computational) Let $De_n = 2^{-n}e_n$ on ℓ^2, where $e_1, e_2, \ldots$ is the standard orthonormal basis ($n \geq 1$). Find $\|D\|$, the adjoint $D^\dagger$, the point spectrum, and the full spectrum; is D self-adjoint? compact?

Answer. $\|D\| = \frac{1}{2}$; $D^\dagger = D$ (self-adjoint); point spectrum $\{2^{-n} : n \geq 1\}$; $\mathrm{spec}(D) = \{2^{-n}\} \cup \{0\}$; compact.

Exercise 14.4 (computational) Find the operator norm, the adjoint, and the spectrum of multiplication by $g(x) = 2x$ on $L^2[0, 1]$. Does it have eigenvalues?

Answer. $\|M\| = 2$; self-adjoint ($M^\dagger = M$); $\mathrm{spec}(M) = [0, 2]$; no eigenvalues.

Exercise 14.5 (computational) Let S be the right shift $S(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$ on ℓ^2. Compute $S^\dagger S$ and $SS^\dagger$, give $\|S\|$, and decide whether S is an isometry and whether it is unitary.

Answer. $S^\dagger S = \mathrm{id}$; $SS^\dagger = \mathrm{id} - |e_1\rangle\langle e_1|$ (projection onto $e_1^\perp$); $\|S\| = 1$; isometry but not unitary.

Exercise 14.6 (computational) For $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on $\mathbb{C}^2$, find the adjoint $A^\dagger$, classify A (self-adjoint / unitary / normal), and give $\mathrm{spec}(A)$.

Answer. $A^\dagger = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; A is unitary (hence normal), not self-adjoint; $\operatorname{spec}(A) = \{i, -i\}$.

Exercise 14.7 (proof) Classify $\lambda = 0$ as point, continuous, or residual spectrum in each case:

1. P is a nonzero orthogonal projection with nontrivial kernel;
2. $De_n = n^{-1}e_n$ on ℓ^2;
3. S is the unilateral right shift on ℓ^2.

For the last two, justify the density or non-density of the range. Then explain why a bounded normal operator cannot have residual spectrum.

Hint. Finite-support sequences lie in range D; range $S = e_1^\perp$. For normal A, compare null A and null $A^\dagger$.

Exercise 14.8 (proof) Show that a bounded diagonal operator $De_n = d_n e_n$ on ℓ^2 (with $\sup_n |d_n| < \infty$) has $\|D\| = \sup_n |d_n|$.

Position, momentum, and the commutator

Exercise 14.9 (proof) On $L^2(\mathbb{R})$ prove both position and momentum are unbounded from normalized inputs.

1. For $X\psi = x\psi$ and $\psi_n = \mathbf{1}_{[n,n+1]}$, prove $\|\psi_n\| = 1$ and compute $\|X\psi_n\|^2$.
2. Fix a normalized $\phi \in \mathcal{S}(\mathbb{R})$ and, for the wave number $k \in \mathbb{R}$, set $\eta_k(x) = e^{ikx}\phi(x)$. Prove η_k is normalized and in $\mathcal{S}(\mathbb{R})$, compute $\hat{p}\eta_k$ for $\hat{p} = -i\hbar\, d/dx$, and use the reverse triangle inequality to conclude that $\hat{p}$ is unbounded.

Answer. $\|X\psi_n\|^2 = n^2 + n + 1/3 \to \infty$; $\hat{p}\eta_k = \hbar k \eta_k + e^{ikx}\hat{p}\phi$ and $\|\hat{p}\eta_k\| \geq \hbar|k| - \|\hat{p}\phi\| \to \infty$.

Exercise 14.10 (proof) On the common invariant domain $\mathcal{S}(\mathbb{R})$, first explain why $\hat{x}\hat{p}\psi$ and $\hat{p}\hat{x}\psi$ are both defined. Then verify $[\hat{x}, \hat{p}]\psi = i\hbar\psi$ for $\hat{p} = -i\hbar\, d/dx$. Finally prove that no finite matrices satisfy this exact CCR and state precisely what the trace obstruction does—and does not—show about quantum models.

Hint. $\mathcal{S}(\mathbb{R})$ is invariant under multiplication by x and differentiation; then take the finite-dimensional trace.

Domain convention for Exercises 14.11–14.16 and 14.18: all displayed identities are equalities on $\mathcal{S}(\mathbb{R})$. This dense space is invariant under $\hat{x}$, $\hat{p}$, and their polynomial products, so every composition used below is defined. These repeated calculations are extension practice rather than part of the selected core route.

Exercise 14.11 (computational) Using $[A, BC] = [A, B]C + B[A, C]$, compute $[\hat{x}, \hat{p}^2]$.

Answer. $[\hat{x}, \hat{p}^2] = 2i\hbar\,\hat{p}$.

Exercise 14.12 (computational) Using $[A, BC] = [A, B]C + B[A, C]$ and $[\hat{x}, \hat{p}] = i\hbar$, compute $[\hat{p}, \hat{x}^2]$.

Answer. $[\hat{p}, \hat{x}^2] = -2i\hbar\,\hat{x}$.

Exercise 14.13 (computational) Compute $[\hat{x}, \hat{p}^3]$.

Answer. $[\hat{x}, \hat{p}^3] = 3i\hbar\,\hat{p}^2$.

Exercise 14.14 (computational) Compute $[\hat{x}^2, \hat{p}^2]$, leaving the answer in terms of $\hat{x}$ and $\hat{p}$.

Answer. $[\hat{x}^2, \hat{p}^2] = 2i\hbar(\hat{x}\hat{p} + \hat{p}\hat{x})$.

Exercise 14.15 (computational) In natural units ($\hbar = 1$, so $[\hat{x}, \hat{p}] = i$) set $a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$ and $a^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p})$. Compute $[a, a^\dagger]$.

Answer. $[a, a^\dagger] = \mathrm{id}$.

Exercise 14.16 (computational) With $a, a^\dagger$ as above and the number operator $N = a^\dagger a$, use $[a, a^\dagger] = \mathrm{id}$ to compute $[N, a]$.

Answer. $[N, a] = -a$.

Exercise 14.17 (conceptual) *(The boundary condition is part of the observable.)* Let $T = -\dfrac{d^2}{dx^2}$ act on twice-differentiable functions on $[0, 1]$ with the L^2 inner product.

1. Integrating by parts twice, show that
$$\langle Tu, v\rangle - \langle u, Tv\rangle = \Big[\overline{u}\, v' - \overline{u'}\, v\Big]_0^1.$$
2. Conclude that T is symmetric ($\langle Tu, v\rangle = \langle u, Tv\rangle$) on the Dirichlet domain $u(0) = u(1) = 0$, and explain why the same operator with *no* boundary condition is not.
3. Explain why this boundary-form calculation proves symmetry but does not by itself prove self-adjointness; what further domain equality is needed?

Exercise 14.18 (proof) Prove $[\hat{x}, \hat{p}^{\,n}] = n\, i\hbar\, \hat{p}^{\,n-1}$ for all $n \geq 1$ by induction.

The spectral theorem and dynamics

Exercise 14.19 (proof) State the compact self-adjoint spectral theorem without assuming separability, including the nonzero eigenvalues, their multiplicities, the kernel, and the finite-dimensional case. Then apply it to $Te_n = n^{-1}e_n$ on ℓ^2: write Tv and $\|Tv\|^2$ for $v = \sum_n c_n e_n$ and classify 0.

Answer. Nonzero eigenvalues are finite/countable, finite-multiplicity, and accumulate only at 0; $\mathcal{H} = \operatorname{null} T \oplus \overline{\operatorname{range} T}$ and the kernel completes the eigenbasis. Here $Tv = \sum c_n e_n/n$, $\|Tv\|^2 = \sum |c_n|^2/n^2$, and 0 is continuous spectrum.

Bridge practice: finite ideal measurement and dynamics. Exercises 14.20–14.25 transfer the core spectral language to finite models; they do not count toward core progress.

Exercise 14.20 (computational) For $H = \frac{\hbar\omega}{2}\sigma_z$, evolve $|\psi\rangle(0) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ to time t.

Answer. $|\psi\rangle(t) = \frac{1}{\sqrt{2}}\big(e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle\big)$.

Exercise 14.21 (computational) Measuring σ_x on $|0\rangle$: give the outcome probabilities and $\langle\sigma_x\rangle$.

Answer. Both outcomes probability $\frac{1}{2}$; $\langle\sigma_x\rangle = 0$.

Exercise 14.22 (computational) Let $Te_n = \frac{1}{n}e_n$ on ℓ^2 (compact, self-adjoint) and $v = 3e_1 + 4e_2$. Compute Tv, $\|v\|^2$, $\|Tv\|^2$, $\langle v, Tv\rangle$, and the Rayleigh quotient $\langle v, Tv\rangle/\|v\|^2$.

Answer. $Tv = 3e_1 + 2e_2$; $\|v\|^2 = 25$; $\|Tv\|^2 = 13$; $\langle v, Tv\rangle = 17$; quotient $\frac{17}{25}$.

Exercise 14.23 (computational) A three-level system has $H|n\rangle = n\varepsilon|n\rangle$ for $n = 0, 1, 2$. Evolve $|\psi\rangle(0) = \frac{1}{\sqrt{3}}(|0\rangle + |1\rangle + |2\rangle)$ to time t.

Answer. $|\psi\rangle(t) = \frac{1}{\sqrt{3}}\big(|0\rangle + e^{-i\varepsilon t/\hbar}|1\rangle + e^{-2i\varepsilon t/\hbar}|2\rangle\big)$.

Exercise 14.24 (computational) Measuring σ_z in $|\psi\rangle = \frac{1}{2}(\sqrt{3}\,|0\rangle + |1\rangle)$: give the outcome probabilities and $\langle\sigma_z\rangle$.

Answer. $P(+1) = \frac{3}{4}$, $P(-1) = \frac{1}{4}$; $\langle\sigma_z\rangle = \frac{1}{2}$.

Exercise 14.25 (computational) Using the spectral decomposition $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, compute $e^{i\alpha\sigma_z}$ for real α.

Answer. $e^{i\alpha\sigma_z} = e^{i\alpha}|0\rangle\langle 0| + e^{-i\alpha}|1\rangle\langle 1| = \cos\alpha\ \mathrm{id} + i\sin\alpha\,\sigma_z$.

Exercise 14.26 (proof) Let A be a bounded self-adjoint operator, ψ normalized, and $\rho = |\psi\rangle\langle\psi|$.

1. Prove $\mathrm{tr}(\rho A) = \langle\psi, A\psi\rangle$.
2. If P is the PVM of A, state the probability of a spectral region Ω and the ideal Lüders post-measurement state, including the nonzero-probability condition.
3. State the scope of this simple quantum dictionary and name the general replacements for states, measurements, and open dynamics.

Hint. Use an ONB whose first vector is ψ; then use $P(\Omega)$ and distinguish the core model from density operators, POVMs/instruments, and channels.

Exercise 14.27 (proof) Let A be a possibly unbounded self-adjoint operator with PVM P and $\mu_\psi(\Omega) = \langle\psi, P(\Omega)\psi\rangle$. State and explain:

1. the domain $D(A)$;
2. the difference between bounded Borel $f(A)$ and unbounded $f(A)$;
3. when $(A - zI)^{-1}$ is the bounded resolvent;
4. the continuity hypothesis in Stone's theorem and the domain of its generator derivative.